DURATION: 4 HOURS

DEVELOP SOFTWARE USING ARTIFICIAL INTELLIGENCE (AI) TOOLS/AI Agents FOR THE FOLLOWING USE STORY

Evaluation Criteria (Read this, Very Important)

Evaluation Criteria

The evaluation process will focus on the quality of each deliverable outlined in this document. It is important to note that assessment goes beyond merely ensuring the software functions as intended. The effectiveness of AI tools and agents, as integrated across every phase of the Software Development Life Cycle, will be a primary consideration.

Every deliverable should be comprehensive, presented in a professional manner, and show clear completion. It is recommended to use AI tools to review all deliverables before finalizing them.

Maintain a comprehensive record of all prompts utilized for the creation of various deliverables. Ensure that you are logged into your tool during use so prompt history remains readily accessible.

Key Areas of Assessment

- **Comprehensive Application of AI**: The use of AI in analysis, design, development, testing, and documentation must be demonstrated for all required outcomes.
- **Impactful Implementation:** Evaluators will look for clear evidence of how AI technologies have positively influenced each stage of the software development process.

- **Deliverable Quality and Completeness**: Each deliverable must be thoroughly developed, meeting all specified requirements with high quality and completeness.
- **Innovative Approach:** Out-of-the-box thinking will be valued, especially in the creation of deliverables. For example, submitting a Functional Specification Document in HTML format instead of the traditional Word document format will be considered innovative.

INDICATIVE LIST OF TOOLS THAT CAN BE USED

AI App & Workflow Builders

- Lovable AI (Free tier)
- Bolt.new (Free credits)
- Replit (Free tier with AI features)
- Vercel v0 (Free tier)

LLM & Agent Development Platforms

- Google AI Studio (Gemini)
- OpenAI Playground (Free credits)
- Azure AI Studio (Free tier)
- Hugging Face Spaces
- LangChain (Open source)
- LlamaIndex (Open source)
- CrewAI (Open source)
- AutoGen (Open source)

AI Coding & Code-Quality Agents

- Codeium (Free)
- Cursor (Free tier)
- Continue.dev (Open source)
- Semgrep (Free tier)
- SonarQube Community Edition

AI for IT – SDLC

AI for Requirements, Documentation & Analysis

- ChatGPT (Free)
- Claude (Free tier)
- Notion AI (Limited free usage)
- Obsidian + community AI plugins

AI for UI / UX & Wireframe Design

- Lovable AI (Free tier) ✅ *(explicitly included)*
- Uizard (Free tier)
- Figma (Free tier with AI features)
- Penpot (Open source)
- Diagrams.net (Draw.io AI)
- Whimsical (Free tier)

AI for Testing & QA

- Postman AI (Free tier)
- Playwright + AI plugins
- Selenium IDE + AI extensions
- OpenTest (Open source)

AI for Architecture, Data Modeling & Diagrams

- DBdiagram.io (Free tier)
- Creately AI (Free tier)
- PlantUML (AI-assisted via prompts)

AI Agent UI / Demo Frameworks

- Streamlit (Open source)

- Gradio (Open source)
- Chainlit (Open source)

AI Search & Research

- Perplexity AI (Free)
- Phind (Free)
- You.com (Free AI modes)

Local / Open-Source Models & Runtimes

- Ollama
- LM Studio
- Mistral (Open models)
- Groq (Free tier)

User Story

AI-Assisted Flexible Work & Life Balance Request System

As a woman employee, I want to request flexible work arrangements—such as working from home, shift changes, and caregiving support—with the assistance of AI. The AI should help me structure my request, track approval status, and monitor outcomes to ensure that work-life balance is promoted fairly and consistently across the organization.

Expanded Details

- **Request Initiation:** Employees can submit requests for flexible work arrangements through an intuitive interface. The AI guides users to specify their needs (e.g., WFH, shift changes, caregiving support) and captures relevant context such as timing, urgency, and supporting information.
- **AI-Generated Justification:** The AI analyzes the employee's informal reasoning, converting it into a professionally formatted justification. It suggests relevant request categories and references applicable company policies, ensuring the submission aligns with organizational guidelines.
- **Approval Workflow:** Requests are routed automatically to designated approvers based on organizational hierarchy and type of request.
- **Policy Tagging:** Each request is tagged with relevant company policies and procedures, helping both employees and managers understand the criteria for approval and ensuring compliance.
- **Outcome Monitoring:** The system tracks outcomes of approved or denied requests, enabling employees to receive feedback and suggestions for future submissions. The AI provides analytics on the effectiveness of flexible arrangements and their impact on work-life balance.

Functional Scope

- **Request Intake:** Guided submission process for flexible work arrangements.
- **AI-Generated Justification: Intelligent conversion of informal reasoning to professional requests**.
- **Approval Workflow:** Automated routing, tracking, and follow-up of requests.
- **Policy Tagging**: Linking requests to company policies for clarity and compliance.
- **Outcome Monitoring:** Feedback and analytics on request outcomes.
- **Reporting dashboard**

Deliverables

All the deliverables are expected to be of production-level quality, comprehensive and ensuring excellence in both content and look & feel.

1. Functional requirements document
2. Application Architecture and Architecture document
3. Database design / Data Models – No JSON input of data, use any RDBMS, deliverables include logical model, physical model and ER diagram
4. UI/Ux/Wireframe Design
5. Software code & Working Software
6. Functional Test scripts in a CSV or Excel
7. Traceability Matrix in CSV or Excel
8. End User manual - Screens and help guide
9. AI Agent – **"Code Quality Checker"** built with an AI framework and Streamlit or Gradio; takes code and checklist as input and provide observations against each checklist item and suggestions for improvements if any

Instructions

Use AI tools or Build Your Own AI Agents (BYOAA) to develop the software and deliverables.

- **Selection of Tools:** Carefully evaluate and select the most suitable AI tools or frameworks for each phase of the project. If building custom AI agents, ensure they are scalable, maintainable. Additional weightage will be given if team builds their own AI agent to create the deliverables.
- **Quality Standards:** Adhere strictly to coding standards and best practices for software development. Apply consistent design patterns, code reviews, and automated testing to maintain high quality.
- **Integration:** Ensure seamless integration of components with the overall application architecture.
- **Testing:** Prepare comprehensive test cases, including both functional and non-functional scenarios.

- **Documentation:** Maintain clear and detailed documentation for all deliverables, including architecture diagrams, data models, and user manuals. Update documents regularly as changes are made.
- **User Experience:** Prioritise a user-centric approach in UI/UX design. Avoid using Streamlit or Gradio for the user interface.